# Attention in Attention for Hyperspectral With High Spatial Resolution ($H^2$) Image Classification

Ge Tang, Xinyu Wang, *Member, IEEE*, Hengwei Zhao, Xin Hu, Guang Jin, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— An inevitable trend of hyperspectral remote sensing has been toward hyperspectral with high spatial resolution ($H^2$) images. However, the higher resolution also brings higher spatial/spectral heterogeneity of surface features, which increases the difficulty of fine classification. Fully using global spatial–spectral features and contextual information is an effective method to alleviate spatial/spectral heterogeneity. Recently, to extract global spatial–spectral features with long-range dependencies, the self-attention mechanism has been widely used in $H^2$ image classification and has achieved excellent results. As is well known, the simultaneous use of spatial and spectral information has always been a key aspect of hyperspectral image (HSI) processing; however, the current spatial and spectral attention modules only focus on the spatial and spectral features separately. This prevents further improvement in network performance, especially when the sample size is small. Therefore, a spatial–spectral attention-in-attention network ($S^2$AiANet) is proposed, which solves the problem of the current spatial–spectral attention maps only focusing on single features through the spatial–spectral attention-in-attention ($S^2$AiA) module. In addition, a multiscale attention (MSA) module is proposed to enhance the network's adaptability to various complex scenarios. The experiments on two $H^2$ datasets and one classic HSI dataset demonstrate that $S^2$AiANet can achieve a significant performance improvement compared with the state-of-the-art hyperspectral classifiers.**

***Index Terms*— Attention in attention, deep learning, hyperspectral image (HSI) classification, multiscale attention.**

## I. Introduction

HYPERSPECTRAL with high spatial resolution ($H^2$) images [1] contain both nanolevel fine spectral information and submeter-level precise spatial information, which makes it possible to use hyperspectral remote sensing for fine-scale ground object recognition. $H^2$ images have been widely applied in geology [2], forestry [3], [4], [5], [6], urban functional zones [7], environmental monitoring… [8], and agriculture [9], [10], [11]. For precision agriculture in particular [12], $H^2$ images play an important role in fine crop classification [13] and distribution mapping [14] and can reach a submeter-level classification accuracy and allow subcategory classification of crops.

The early hyperspectral image (HSI) classification method [15], [16], [17], [18] features were handcrafted, and the classification results were particularly dependent on the prior information about the images and the experience of experts. Recently, data-driven deep learning technology has been widely used in HSI classification [19], especially convolutional neural network (CNN)-based methods. These methods can automatically learn deep spectral–spatial features that are most suitable for classification. To use the spatial and spectral information in HSI adequately, these CNN-based methods typically follow a network framework called patch-based [20]. In brief, they need to select a definite patch with a fixed size of $N \times N$, which centers on the pixel about to be classified. The patch is input and the result of the classification is assigned to the central pixel. The patch-based HSI classification methods can extract spatial and spectral features simultaneously. As a result, they can obtain the better classification results than the methods that just use spectral information.

The serious spatial heterogeneity and spectral variability of $H^2$ images increases the intraclass variance of ground objects and decreases the interclass distance [21], and it places higher demands on the performance of the network. The patch-based methods can only use local spatial–spectral features and are unable to accurately classify $H^2$ images. To solve the above problems, fully convolutional network (FCN)-based methods [22] have been used in hyperspectral classification networks to consider the global spatial–spectral features. FCN-based models have been successfully used for HSI classification through global stochastic stratified ($GS^2$) sampling [23]. These methods take the entire image as input, providing a foundation for the network to utilize the global contextual information, and have been shown to be able to achieve superior accuracy in HSI classification [22], [23]. However, the receptive field of CNN is limited and cannot capture the long-range dependencies [24] between global pixels, making the network unable to accurately classify $H^2$ images.

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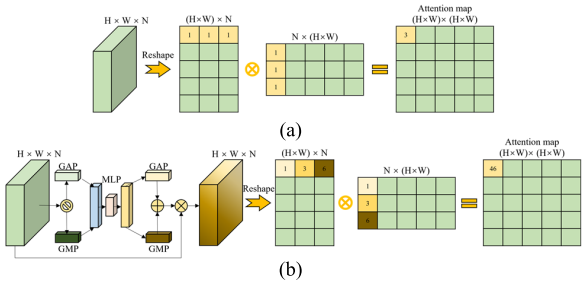


Fig. 1. Difference of the different spatial attention maps. (a) Only consider spatial information. (a) Consider spatial and spectral information.

Therefore, attention modules [25], which can assign different weights to pixels at different positions and bands in different channels, have been widely used in HSI classification. The attention mechanism, which comes from the human visual system, can help the network pay more attention to the features that are helpful for classification and ignore the others [26]. Many networks based on attention mechanisms or incorporating attention modules have been shown to be able to achieve superior accuracy in HSI classification [27], [28], [29], [30], [31], [32]. In addition, in the face of the problem that long-range dependencies cannot be captured only by CNN operation, recently, the nonlocal modules [33] based on a self-attention mechanism have been widely used. These modules can enhance the ability of the network to extract global features and capture the long-range dependencies, to further improve the performance [34], [35]. Specific improvements have also been made for hyperspectral data, including capturing long-range spatial–spectral dependencies [36], [37]; full context feature extraction for spatial, spectral, and scale information [38]; and the use of cross-attention to alleviate the massive computational complexity [39].

There are many networks with attention mechanisms that can accurately classify $H^2$ images and obtain excellent results. However, the current methods still face two problems.

1) The current spatial and spectral attention mechanisms only focus on the spatial or spectral features separately because the weight of each band is the same when calculating the global spatial attention map and the weight of each position is the same when calculating the global spectral attention map, which is clearly unreasonable. This means that the network cannot calculate the most effective spatial attention map by fully evaluating the importance of spectral information in different bands, and it also means that the network cannot calculate the most effective spectral attention map by fully evaluating spatial information in different positions. Therefore, in the current self-attention mechanism, the calculation method of multiplying and adding corresponding bands between different pixels and multiplying and adding corresponding pixels between different bands is not reasonable enough, which suppresses any further performance improvement. Especially when the number of training samples is extremely reduced, the classification accuracy of the network will be significantly reduced. Fig. 1 shows the difference between spatial attention maps that only consider spatial information and those that simultaneously consider spatial and spectral information.

2) The current multiscale features module does not fully consider the weights between different scale features. Actually, for different types of crops, such as rape and cabbage, the importance of features at different scales is different. Therefore, how to make the model fully consider the weight relationship between different scale features and improve the network's ability to utilize different scale features is an important issue currently faced by multiscale modules.

In this article, a spatial–spectral attention-in-attention ($S^2$AiA) module is proposed for fully considering the spatial–spectral information at the same time. The $S^2$AiA module can fully consider the influence of different bands on the spatial attention map and the influence of different positions on the spectral attention map. Improving the independent correlation calculation in the attention mechanism can solve the problem of noise and ambiguous weights in the attention map while extracting more effective spatial attention and spectral attention maps. In addition, to enhance the network's adaptability to various complex scenarios, a multiscale attention (MSA) module is proposed, which explicitly distinguishes the weights of the features at different scales. The MSA module allows the network to extract more robust scale features when faced with images with significant differences in scale between different scenes or ground objects and significantly reduces the misclassification at the boundaries of different-sized parcels in the classification map. Finally, the S2AiA network ($S^2$AiANet) based on an encoder–decoder structure is proposed for the precise classification of $H^2$ remote sensing images with limited training samples.

This article is organized as follows. Section II briefly reviews patch-based/FCN-based methods and attention mechanisms in HSI classification. Section III then elaborates the specific content of the $S^2$AiANet. The extensive experiments and analyses are provided in Section IV. Finally, Section V concludes this article.

## II. Related Work

### A. Patch-Based Classification Methods

One mainstream current HSI classification method of deep learning follows the patch-based local image input method. The patch-based methods typically take the pixel to be classified as the center and select its surrounding $N \times N$ sized space block as input to achieve the network training. In this way, with the input changed from a single pixel to a patch, spatial and spectral features can be simultaneously extracted by the network for HSI classification. Some classic patch-based methods have been proposed and achieved great results. For example, R-PCACNN [40] uses a 2-D CNN to encode the spatial and spectral information, HybridSN [41] uses a 3-D CNN to further utilize spatial–spectral joint information, and SSAN [29] builds a two-branch network to extract the spatial and spectral features independently. Despite the patch-based classification methods already obtaining superior classification results, they still cannot use global spatial and spectral information. Because these methods' inputs are definite patches with a fixed size of $N \times N$, they are unable to utilize the global spatial–spectral features. This causes the

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network to generate many isolated misclassification regions when faced with images with extremely high spatial–spectral heterogeneity. This prediction method also causes the network to spend a large amount of computing time when outputting the classification results.

### B. FCN-Based Classification Methods

Another mainstream modern method of deep learning follows the FCN-based global image input method. To solve the problems of the patch-based frameworks, researchers have considered the network with an FCN-based encoder–decoder because it can use the whole image as input and perform end-to-end prediction. The FCN structure [42] was originally proposed for semantic segmentation tasks and has been shown to have an excellent performance in computer vision tasks. Among them, the structure of U-Net [43], which consists of the encoder–decoder and lateral connections, is most widely used. Similarly, the purpose of HSI classification is to assign a unique label to each pixel, which is the same as the purpose of the semantic segmentation tasks. Therefore, an idea is to migrate models in the field of semantic segmentation to HSI classification tasks. However, it is difficult because their training samples are labeled differently. The former is densely labeled pixels, while the latter is highly sparse discrete labeled pixels. To solve the problem of network training, $GS^2$ sampling has been proposed, which can obtain diverse gradients and ensure that the network can be trained. Based on this, the patch-free global learning framework (FPGA [23]) is proposed. FPGA can take the whole image as the input to provide the basis for the network to extract global spatial–spectral information while saving a large amount of computational coast and time during the network prediction and inference.

### C. Attention Mechanisms in HSI Classification

To enable networks to distinguish features of different importance levels, attention mechanisms have been widely applied in HSI classification tasks. Attention mechanisms come from the excellent processing ability of the human visual system for a large amount of information. In human daily life, the visual system continuously receives a lot of scene information [44]. However, a human being's ability to process information is limited. Therefore, after receiving the light stimulation signals, the visual system suppresses unimportant regional features while highlighting the features of the regions of interest. This is an important means for humans to quickly extract the most valuable information from a large amount of information through limited processing resources [45]. The attention mechanism approach is also very effective in HSI classification [25]. On this basis, a simple network architecture based solely on attention mechanisms was proposed by Vaswani et al. [33] to improve the network performance. The nonlocal feature extraction model can effectively help a network establish long-distance dependencies. Subsequently, a variety of nonlocal network extensions have been introduced, such as the dual-attention network [35], which extracts nonlocal features of the spatial position and spectral dimension simultaneously. Another example is the criss-cross attention network [46], which uses criss-cross attention to achieve the purpose of global information interaction while improving the efficiency of the network. However, in the current self-attention mechanism, the independent correlation calculations between different pixels and bands without considering weights can prevent further improvement in network performance.

Therefore, in this study, our motivation was to develop an attention-in-attention module to solve the problem of independent correlation computation, make the calculation of the attention map more reasonable, and further improve the network performance.

## III. Proposed Method

To allow the network to use global information and achieve fast prediction, the proposed network architecture uses an FCN as the backbone network. There are two main parts in the proposed network: 1) the $S^2$AiA module for global spatial–spectral context information and 2) the MSA module for different scales feature extraction. These modules are used to enhance the network's attention to global features and its adaptability in different scenarios. The details of each part are described in the following.

### A. Network Architecture

The architecture of the proposed $S^2$AiANet network is shown in Fig. 2. The previous patch-based methods are abandoned because they can only obtain the local spatial–spectral information in a fixed area around pixels to be classified. We instead use an FCN-based method, which uses the whole image as input during the training and predicting process of the network, and the inference only needs to be run once to obtain the classification results of the entire image. $S^2$AiANet includes three parts: an encoder, a decoder, and lateral connections. In the encoder, a $1 \times 1$ convolution is used to fix the number of feature channels for input images, facilitating subsequent unified processing, and then, the $3 \times 3$ convolution and max-pooling operations are alternately used to extract deep features. At the deepest level, the $S^2$AiA module and the MSA module are, respectively, used to enable the network to use global spatial–spectral context information and multiscale features, to improve the classification accuracy. In the decoder, the $3 \times 3$ convolution and $2\times$ upsampling operations are used to restore the image size layer by layer. In the lateral connections, $1 \times 1$ convolution is used to fuse the feature maps of the same size in the corresponding encoder, to more accurately restore the spatial details. Finally, we use a size of $1 \times 1$ convolutional layer to specify the number of feature channels as the number of categories, and uses softmax operation to predict the category labels of each pixel.

### B. $S^2$AiA Module

The mainstream nonlocal operation modules are now based on self-attention. The spatial/spectral attention map is obtained and then applied to the original feature map to make the network to establish dependence relationships between global

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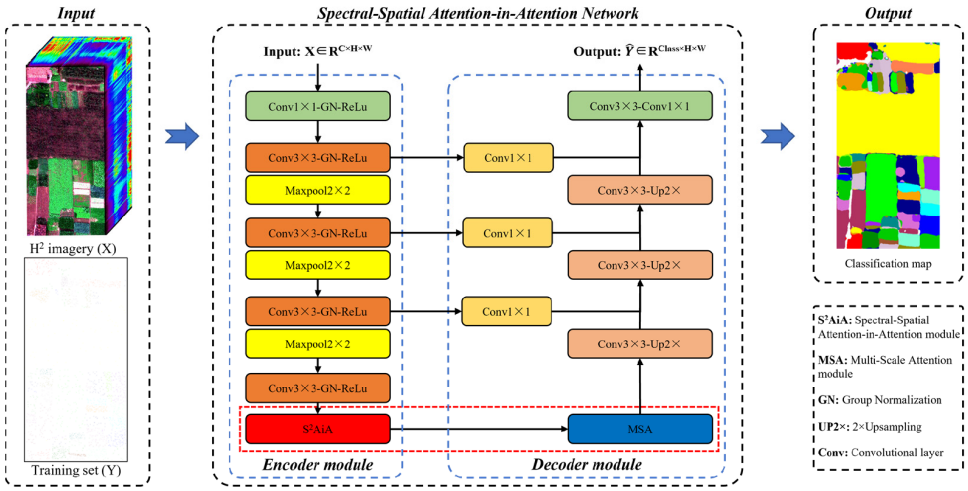


Fig. 2. Architecture of S$^2$AiANet for H$^2$ image classification.

pixels, thereby utilizing global spatial–spectral context information. We take the calculation of global spatial context information extraction as an example. To obtain the spatial attention map, we reshape the feature map $R^{C\times H\times W}$ to $R^{HW\times C}$ and $R^{C\times HW}$ and then perform a matrix operation. Each pixel interacts with all the other pixels to obtain the spatial context information between them. This process can be expressed as

$$M^{HW\times HW} = f\left(g\left(R^{C\times H\times W}\right)\right)\cdot f\left(g\left(R^{C\times H\times W}\right)\right) \quad (1)$$

where $M^{HW\times HW}$ represents the spatial attention map, $f(\cdot)$ is the reshape operation (from $C\times H\times W$ to $HW\times C$ and $C\times HW$), and $g(\cdot)$ is the $1\times 1$ convolution operation. The spatial attention map is then applied to the original feature map, and the global spatial context information is extracted. This process can be expressed as

$$R_{\text{spa}}^{C\times H\times W} = f\left(f\left(g\left(R^{C\times H\times W}\right)\right)\cdot M^{HW\times HW}\right) + R^{C\times H\times W} \quad (2)$$

where $R_{\text{spa}}^{C\times H\times W}$ represents the output with global spatial context information, $f(\cdot)$ is the reshape operation (from $C\times H\times W$ to $C\times HW$), and $g(\cdot)$ is the $3\times 3$ convolution operation. Similarly, the process of global spectral context information extraction can be expressed as

$$M^{C\times C} = f\left(g\left(R^{C\times H\times W}\right)\right)\cdot f\left(g\left(R^{C\times H\times W}\right)\right) \quad (3)$$

where $M^{C\times C}$ represents the spectral attention map, $f(\cdot)$ is the reshape operation (from $C\times H\times W$ to $C\times HW$ and $HW\times C$), and $g(\cdot)$ is the $1\times 1$ convolution operation. The spectral attention map is then applied to the original feature map, and the global spectral context information is extracted. This process can be expressed as

$$R_{\text{spe}}^{C\times H\times W} = f\left(M^{C\times C}\cdot f\left(g\left(R^{C\times H\times W}\right)\right)\right) + R^{C\times H\times W} \quad (4)$$

where $R_{\text{spe}}^{C\times H\times W}$ represents the output with global spectral context information, $f(\cdot)$ is the reshape operation (from $C\times H\times W$ to $HW\times C$), and $g(\cdot)$ is the $3\times 3$ convolution operation.

However, the calculation of the attention map is equal in weight. We take the calculation of the global spatial context information extraction as an example. In this process, we assume that the weights of the different bands of a single pixel are the same, and there is no information interaction between different bands. Clearly, this is unreasonable. First, in the current self-attention mechanism, the independent correlation calculations between different pixels and bands without considering weights can prevent further improvement in network performance. Second, to obtain the most effective spatial attention map, when calculating the spatial attention map, the weights of the different bands should be different. The details of the differences between the traditional calculation method and our proposed calculation method are shown in Fig. 1. Similarly, when calculating the band attention map, the weights of the different positions should be different.

Therefore, the S$^2$AiA module is proposed, by adding a corresponding spatial or spectral attention module before the calculation of the spatial or spectral self-attention feature map, to solve the above problems. Fig. 3 shows the details of the S$^2$AiA module. To extract long-range spatial–spectral context information more reasonably, the spectral attention module should be used in front of the spatial attention map, and the spatial attention module should be used in front of the band attention map. The process of the S$^2$AiA module can be expressed as

$$M^{HW\times HW} = f\left(g\left(\theta\left(R^{C\times H\times W}\right)\right)\right)\cdot f\left(g\left(\theta\left(R^{C\times H\times W}\right)\right)\right) \quad (5)$$

$$M^{C\times C} = f\left(g\left(\psi\left(R^{C\times H\times W}\right)\right)\right)\cdot f\left(g\left(\psi\left(R^{C\times H\times W}\right)\right)\right) \quad (6)$$

where $\theta(\cdot)$ represents spectral attention, $\psi(\cdot)$ is spatial attention, $g(\cdot)$ represents the $1\times 1$ convolution operation, $f(\cdot)$ is the reshape operation (spatial branch: from $C\times H\times W$ to $HW\times C$ and $C\times HW$; spectral branch: from $C\times H\times W$ to $C\times HW$ and $HW\times C$)

$$R_{\text{spa}}^{C\times H\times W} = f\left(f\left(g\left(R^{C\times H\times W}\right)\right)\cdot M^{HW\times HW}\right) + R^{C\times H\times W} \quad (7)$$

$$R_{\text{spe}}^{C\times H\times W} = f\left(M^{C\times C}\cdot f\left(g\left(R^{C\times H\times W}\right)\right)\right) + R^{C\times H\times W} \quad (8)$$

$$R_{\text{spaspe}}^{2C\times H\times W} = F\left(R_{\text{spa}}^{C\times H\times W}, R_{\text{spe}}^{C\times H\times W}\right) \quad (9)$$

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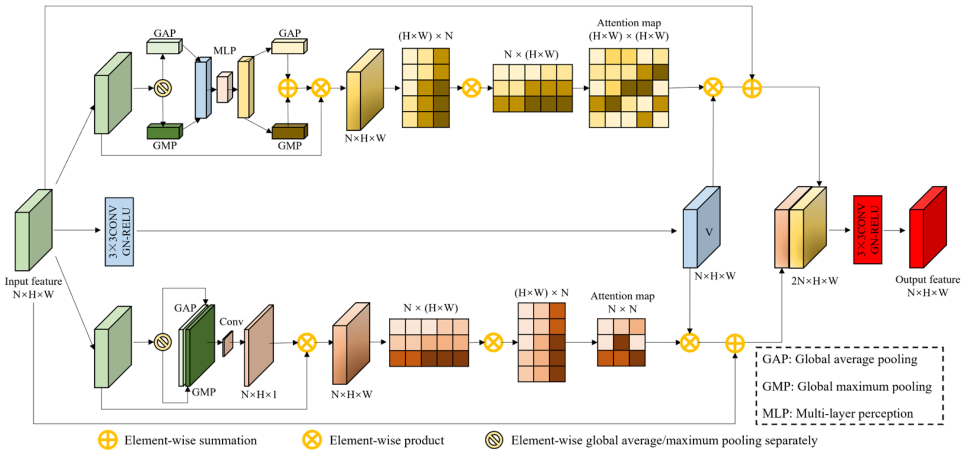


Fig. 3. Details of S²AiA module. $N$, $H$, and $W$ represent the band number, height, and width, respectively.

where $R_{\text{spa}}^{C\times H\times W}$ represents the output with global spatial context information, $R_{\text{spe}}^{C\times H\times W}$ is the output with global spectral context information, and $R_{\text{spaspe}}^{2C\times H\times W}$ is the output with global spatialspectral context information. $F(\cdot)$ represents the concatenation operation

$$R_{\text{out}}^{C\times H\times W} = g\left(R_{\text{spaspe}}^{2C\times H\times W}\right) \tag{10}$$

where $R_{\text{out}}^{C\times H\times W}$ represents the output extracted by S²AiA and $g(\cdot)$ is the $3\times 3$ convolution operation. The number of characteristic channels is the same as the input so that the module can be more easily added to the network.

It is worth noting that we have placed the S²AiA module at the deepest layer of the network. Proverbially, the computation cost of extracting long-range contextual information based on self-attention mechanism is greatly affected by the size of the feature map. Also, on this basis, module S²AiA module uses the attention-in-attention method to focus on spatial and spectral information separately through parallel connection. It will inevitably increase the computation cost of the network. Therefore, to balance the performance and computation cost of the network, the S²AiA module is placed at the deepest layer for processing the small-size feature maps after multiple pooling operations.

### C. MSA Module

To more accurately classify different crops in different planting areas, some researchers have also begun to pay attention to the ability of HSI classification networks to extract multiscale features [47]. Furthermore, as the influence of different-scale features on the classification results is different, S³ANet [30] introduced a scale attention mechanism to give different weights to different-scale features. However, S³ANet uses an additive calculation method to make the following attention more similar to channel attention. We have improved on this and developed the MSA module. Fig. 4 shows the details of the MSA module. By using "concatenate" instead of "add," this reduces the computational cost and makes the following attention act on the feature channels of different scales directly.

As shown in Fig. 4, to extract different scale features, convolutions of different receptive fields are used. Also, based

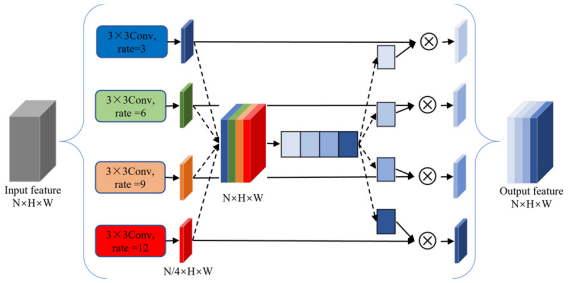


Fig. 4. Details of the MSA module. $N$, $H$, and $W$ represent the band number, height, and width, respectively.

on experience, four atrous convolutions with rate $= 3, 6, 9$, and 12 are, respectively, applied to the MSA module to extract the multiscale features $X_1, X_2, X_3, X_4 \in R^{N/4\times H\times W}$ from the input feature map $X \in R^{N\times H\times W}$. The feature maps are then concatenated to produce the multiscale feature map $X \in R_{\text{MS}}^{N\times H\times W}$. The next MSA module can be expressed as

$$R_{\text{MSA}}^{N\times H\times W} = F_{i=1}^{4}\left(f_{\text{softmax}}\left(f_{\text{Fc}}\left(f_{\text{Gap}}\left(R_{\text{MS}}^{N\times H\times W}\right)\right)\right)\right) \tag{11}$$

where $f_{\text{Gap}}(\cdot)$ represents the global average pooling, $f_{\text{Fc}}(\cdot)$ is a fully connected layer, $f_{\text{softmax}}(\cdot)$ is the softmax operation, and $F_{i=1}^{4}(\cdot)$ means concatenating the four different-scale feature maps.

## IV. Experiments

In this section, including two H² remote sensing image datasets [1] with a nanometer spectral resolution and a centimeter spatial resolution: the WHU-Hi-LongKou and WHU-Hi-HongHu and one classic hyperspectral dataset, the Pavia University is conducted to verify the effectiveness and generalization of the proposed S²AiANet. Also, through a series of experimental analysis and ablation experiments, it is proven that the proposed method has achieved further performance improvement.

### A. Datasets

*1) WHU-Hi-LongKou:* The WHU-Hi-LongKou dataset was obtained by an 8-mm focal length Headwall Nano-Hyperspec imaging sensor in Longkou, Hubei, China, in 2018. It is composed of 550 × 400 pixels, including 270 bands from 400 to

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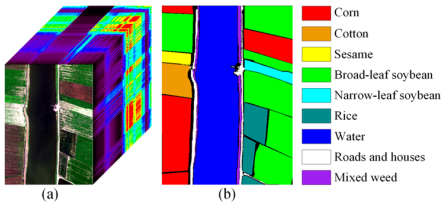
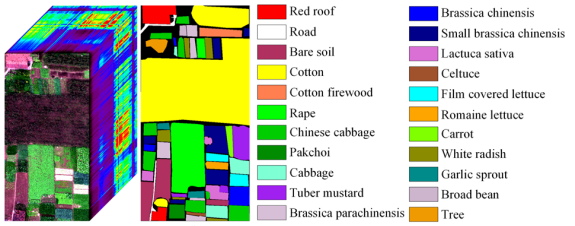



(a) (b)

| No. | Class name | Train | Test | Samples |
|---|---|---|---|---|
| C1 | Corn | 25 | 34486 | 34511 |
| C2 | Cotton | 25 | 8349 | 8374 |
| C3 | Sesame | 25 | 3006 | 3031 |
| C4 | Broad-leaf soybean | 25 | 63187 | 63212 |
| C5 | Narrow-leaf soybean | 25 | 4126 | 4151 |
| C6 | Rice | 25 | 11829 | 11854 |
| C7 | Water | 25 | 67031 | 67056 |
| C8 | Roads and houses | 25 | 7099 | 7124 |
| C9 | Mixed weed | 25 | 5204 | 5229 |

(c)

Fig. 5. WHU-Hi-LongKou dataset. (a) Image cube. (b) Ground-truth image. (c) Category and sample settings.




(a) (b)

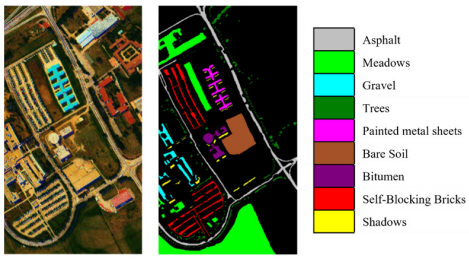
| No. | Class name | Train | Test | Samples |
|---|---|---|---|---|
| C1 | Red roof | 25 | 14016 | 14041 |
| C2 | Road | 25 | 3487 | 3512 |
| C3 | Bare soil | 25 | 21796 | 21821 |
| C4 | Cotton | 25 | 163260 | 163285 |
| C5 | Cotton firewood | 25 | 6193 | 6218 |
| C6 | Rape | 25 | 44532 | 44557 |
| C7 | Chinese cabbage | 25 | 24078 | 24103 |
| C8 | Pakchoi | 25 | 4029 | 4054 |
| C9 | Cabbage | 25 | 10794 | 10819 |
| C10 | Tuber mustard | 25 | 12369 | 12394 |
| C11 | Brassica parachinensis | 25 | 10990 | 11015 |
| C12 | Brassica chinensis | 25 | 8929 | 8954 |
| C13 | Small Brassica chinensis | 25 | 22482 | 22507 |
| C14 | Lactuca sativa | 25 | 7331 | 7356 |
| C15 | Celtuce | 25 | 977 | 1002 |
| C16 | Film covered lettuce | 25 | 7237 | 7262 |
| C17 | Romaine lettuce | 25 | 2985 | 3010 |
| C18 | Carrot | 25 | 3192 | 3217 |
| C19 | White radish | 25 | 8687 | 8712 |
| C20 | Garlic sprout | 25 | 3461 | 3486 |
| C21 | Broad bean | 25 | 1303 | 1328 |
| C22 | Tree | 25 | 4015 | 4040 |

(c)

Fig. 6. WHU-Hi- HongHu dataset. (a) Image cube. (b) Ground-truth image. (c) Category and sample settings.

1000 nm, and the image spatial resolution is 0.463 m. The research area consists of nine categories, including six types of crops, which is a simple agricultural scene. The specific information of this dataset is shown in Fig. 5, including the ground truth and the configuration of the training and test samples.

*2) WHU-Hi-HongHu:* The WHU-Hi-HongHu dataset was obtained by a 17-mm focal length Headwall Nano-Hyperspec imaging sensor in Honghu, Hubei, China, in 2017. It is composed of 940 × 475 pixels, including 270 bands from 400 to 1000 nm, and the image spatial resolution is 0.043 m. The study area consists of 22 categories, including different varieties of the same crop, such as Chinese cabbage/cabbage. The specific information of this dataset is shown in Fig. 6, including the ground truth and the configuration of the training and test samples.




(a) (b)

| No. | Class name | Train | Test | Samples |
|---|---|---|---|---|
| C1 | Asphalt | 25 | 6606 | 6631 |
| C2 | Meadows | 25 | 18624 | 18649 |
| C3 | Gravel | 25 | 2074 | 2099 |
| C4 | Trees | 25 | 3039 | 3064 |
| C5 | Painted metal sheets | 25 | 1320 | 1345 |
| C6 | Bare Soil | 25 | 5004 | 5029 |
| C7 | Bitumen | 25 | 1305 | 1330 |
| C8 | Self-Blocking Bricks | 25 | 3657 | 3682 |
| C9 | Shadows | 25 | 922 | 947 |

(c)

Fig. 7. Pavia University dataset. (a) False-color composite. (b) Ground-truth image. (c) Category and sample settings.

*3) Pavia University:* The Pavia University dataset was obtained by the Reflective Optics System Imaging Spectrometer at Pavia University in northern Italy, in 2001. It is composed of 610 × 340 pixels, including 103 bands from 430 to 860 nm, and the image spatial resolution is 1.3 m. The research area consists of nine categories, which is a common urban scene. The specific information of this dataset is shown in Fig. 7, including the ground truth and the configuration of the training and test samples.

### *B. Experimental Settings*

The proposed S$^2$AiANet was compared with the following classifiers: the classic method of support vector machine (SVM); the patch-based methods of R-PCACNN [40], HybridSN [41], and SSFTT [48] with tokenization transformer; and the FCN-based methods of FPGA [23], S$^3$ANet [30], and FullyContNet [38]. The parameters of these comparison methods were set according to their original papers. To better evaluate the performance of the network, we randomly selected 25 labeled pixels from each class for training on three datasets. The other labeled pixels were used for testing. The stochastic gradient descent (SGD) approach was used for the model training. The initial learning rate was set to 0.005, the weight decay was 0.001, and the momentum was 0.9. The number of learning iterations was 1500. The hyperparameter settings for the standard S$^2$AiANet are listed in Table I, such as the specific input–output of each layer and the convolutional layer settings. All the experiments were conducted five times. All the accuracies were evaluated on the test set, and we choose the best model as the result. In the experiments, in addition to three classic evaluation indicators—the overall accuracy (OA), average accuracy (AA), and the kappa coefficient (kappa), the per-class accuracy was also calculated for comprehensive evaluation of classification performance.

### *C. WHU-Hi-LongKou Dataset Experiment*

The visual classification maps and the accuracy table for the WHU-Hi-LongKou are shown in Fig. 8 and Table II,

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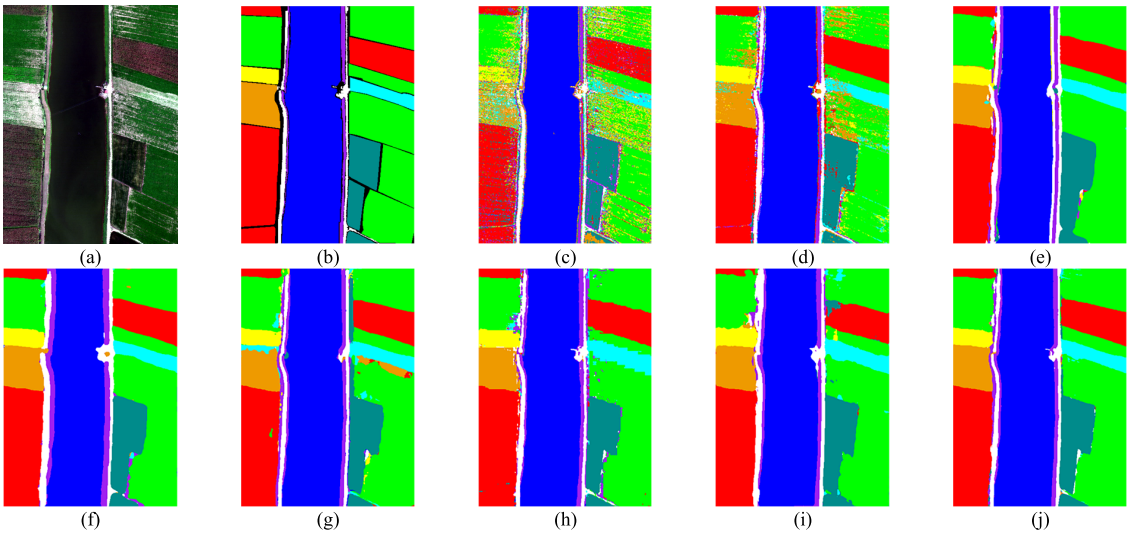

Fig. 8. Classification results for the WHU-Hi-LongKou dataset. (a) False-color image. (b) Ground-truth image. (c) SVM (OA = 92.22%). (d) R-PCACNN (OA = 92.61%). (e) HybridSN (OA = 96.36%). (f) SSFTT (OA = 96.51%). (g) FPGA (OA = 96.06%). (h) S$^3$ANet (OA = 97.08%). (i) FullyContNet (OA = 96.64%). (j) S$^2$AiANet (OA = 99.07%).

TABLE I
CONFIGURATION DETAILS FOR THE STANDARD S $^2$AiANet

| Stage | Block | Network parameters |
|---|---|---|
| Encoder | Stem | 3 × 3 conv, 96 |
| | Block #1 | 3 × 3 conv, 96<br>Maxpool, stride=2 |
| | Block #2 | 3 × 3 conv, 128<br>Maxpool, stride=2 |
| | Block #3 | 3 × 3 conv, 192<br>Maxpool, stride=2 |
| | Block #4 | 3 × 3 conv, 256 |
| Lateral | Latera 4-1 | 1 × 1 conv, 128 |
| | Latera 3-2 | 1 × 1 conv, 128 |
| | Latera 2-3 | 1 × 1 conv, 128 |
| | Latera 1-head | S$^2$AiA, 256<br>MSA, 256 |
| Decoder | Block #1 | 3 × 3 conv, 128<br>Upsample, 2 |
| | Block #2 | 3 × 3 conv, 128<br>Upsample, 2 |
| | Block #3 | 3 × 3 conv, 128<br>Upsample, 2 |
| | Head | 3 × 3 conv, 128<br>1 × 1 conv, class |

respectively. Fig. 8(c)–(j) specifically shows the visual effects of each method on the WHU-Hi-LongKou. The SVM method, which just extracts spectral features with a single pixel as input, shows an obvious granularity phenomenon in the classification map. The patch-based methods are limited by the input size and can only extract local spatial–spectral features. Therefore, even though the WHU-Hi-LongKou is simple and has only nine categories, R-PCACNN still shows a small amount of misclassification, especially when the number of samples is very small. HybridSN and SSFTT are also patch-based methods, but the former uses 3DCNN to enhance feature extraction capabilities and the latter uses a transformer to more effectively extract the long-range context information of the entire patch, reducing the misclassification significantly. The FCN-based methods can use the global spatial–spectral information, and the visual effect of their classification maps is clearly better than the above methods. Similarly, when the training sample size is small, there is still the phenomenon of misclassification at the boundary of the different land-cover regions and there are some easily mixed categories with strong spectral similarity, such as broad-leaf soybean and narrow-leaf soybean. Overall, the classification map generated by the proposed S$^2$AiANet method shows significant improvement in the above two issues.

Table II provides the quantitative classification accuracy of S$^2$AiANet and the comparison methods for the WHU-Hi-LongKou. Because the scene is relatively simple and the spectral variability and spatial heterogeneity are relatively low, the mainstream deep learning methods can achieve a good result (i.e., the OA is more than 92%). Among the different methods, the S$^2$AiANet achieves the best accuracy of 99.07%. Finally, from the above results and analysis, the S$^2$AiANet obtains the best classification result, and it can achieve a high classification accuracy.

### D. WHU-Hi-HongHu Dataset Experiment

Similarly, the visual results are shown in Fig. 9(c)–(j), which specifically demonstrate the visual effects of each method on the WHU-Hi-HongHu, and the classification result accuracy table is provided in Table III. From the visual results, the SVM method is still the worst. The patch-based methods of R-PCACNN have poor classification map performance, with many isolated wrong classification areas. The classification map effect of HybridSN and SSFTT are significantly improved because they consider the spatial–spectral joint feature and context information within the patch with local patches as input, but there are still mistakes, such as classifying cotton as cotton firewood and Brassica chinensis as Chinese cabbage. The classification maps of the FCN-based methods are better. However, they also have a few wrong classification areas, such as classifying cotton as cotton firewood and rape as tuber mustard. The classification map of the proposed S$^2$AiANet method correctly classifies the rape area, and the area of cotton

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TABLE II
CLASSIFICATION ACCURACIES (%) FOR THE WHU-Hi- LongKou Dataset

| No. | SVM | R-PCACNN | HybridSN | SSFTT | FPGA | S$^3$ANet | FullyContNet | S$^2$AiANet |
|---|---|---|---|---|---|---|---|---|
| C1 | 87.15 | 99.88 | 99.72 | 99.98 | 97.83 | 99.59 | 97.91 | 99.80 |
| C2 | 82.52 | 94.20 | 95.45 | 98.10 | 87.89 | 94.89 | 99.02 | 99.44 |
| C3 | 89.19 | 98.74 | 99.23 | 99.86 | 98.90 | 97.60 | 100.00 | 99.67 |
| C4 | 88.85 | 80.76 | 97.55 | 97.14 | 91.95 | 92.75 | 92.96 | 98.03 |
| C5 | 89.89 | 88.71 | 99.3 | 98.69 | 99.00 | 98.64 | 99.87 | 100.00 |
| C6 | 96.30 | 96.01 | 99.86 | 99.95 | 98.22 | 97.11 | 99.63 | 99.89 |
| C7 | 99.94 | 99.59 | 94.07 | 93.60 | 99.74 | 99.89 | 98.49 | 99.82 |
| C8 | 87.14 | 93.94 | 86.05 | 91.09 | 91.94 | 97.82 | 93.88 | 95.46 |
| C9 | 84.20 | 85.74 | 93.12 | 96.65 | 96.84 | 97.91 | 97.88 | 98.58 |
| OA | 92.22 | 92.61 | 96.36 | 96.51 | 96.06 | 97.08 | 96.64 | **99.07** |
| AA | 89.46 | 93.06 | 96.04 | 97.23 | 95.81 | 97.35 | 97.74 | **98.96** |
| Kappa | 89.90 | 0.9047 | 0.9527 | 0.9546 | 0.9487 | 0.9620 | 0.9563 | **0.9878** |

field, which is misclassified as cotton firewood, is smaller. Overall, the proposed S$^2$AiANet method obtains the best classification map.

Table III provides the quantitative classification accuracy of S$^2$AiANet and the comparison methods. There are 22 categories in the dataset, including 18 categories of crops, and the crops are grown in areas of varying sizes, with a large number of broken plots. There are also some similar types of crops, such as cotton and cotton firewood, making the image difficult to classify. The patch-based methods cannot solve the problem of high spatial–spectral heterogeneity. Even SSFTT, which uses a transformer to extract global features, achieves only 90.31%, limited by the input patch size. The accuracies of the FCN-based methods are all more than 91.79%. On this basis, the accuracies of S$^3$ANet, FullyContNet, and S$^2$AiANet, which focus on multiscale and contextual information, show further improvements. The proposed S$^2$AiANet method achieves the best accuracy of 94.94%. Overall, it is not difficult to find that the S$^2$AiANet achieves the best results, in terms of both visual effect and classification accuracy.

### *E. Generalization Experiment on the Pavia University Dataset*

To investigate the generality of the S$^2$AiANet, the classic dataset Pavia University is selected. The visual classification maps and classification result accuracy table for Pavia University are shown in Fig. 10 and Table IV, respectively. From the visual result map, it can be found that the SVM method has the worst classification performance. The patch-based methods of R-PCACNN and HybridSN consider the spatial information, so their classification map effect is clearly improved. Similarly, SSFTT with tokenization transformer performs best among the patch-based methods. However, with the limited receptive fields, the classification maps have a strong sense of granularity. The FCN-based methods achieve the best visual effect. However, FPGA does not consider global context information, resulting in misclassification in the classification map. For example, a part of the meadows in the middle area is misclassified as bare soil. S$^3$ANet uses spatial attention to enhance the spatial information in the shallow feature maps, which in this small sample situation, increases the fragmentation of the classification map instead. FullyContNet considers the full context information, making the classification map too smooth, as can be seen for the self-blocking bricks and gravel. Overall, the proposed S$^2$AiANet method obtains the best classification map.

Table IV provides the quantitative classification accuracy of S$^2$AiANet and the comparison methods for Pavia University. It is not difficult to see that both traditional method and patch-based methods have lower accuracy than FCN-based methods. However, FPGA cannot utilize global context information directly, and the attention modules of S$^3$ANet are unable to extract discriminative generic features and instead extract some nonrepresentative features, in the case of few samples. As a result, its accuracy is not significantly improved. Compared with FullyContNet, which also uses a self-attention mechanism to extract global context information, the proposed S$^2$AiANet method improves the calculation of the self-attention mechanism and achieves superior classification accuracy. In summary, in addition to H$^2$ datasets, the proposed S$^2$AiANet method can still obtain the best classification performance in face of traditional hyperspectral datasets.

### *F. Ablation Experiments*

In this section, the effectiveness of the S$^2$AiA module and MSA module is analyzed. We describe the ablation experiments conducted in this study. For the baseline, we selected the FreeNet of FPGA and used $1 \times 1$ convolution in the deepest layer of the network. We then replaced the baseline with the MSA module, the S$^2$AiA module, and the two modules at the same time.

Compared with the performance of the baseline (FreeNet), the network with the MSA module or S$^2$AiA module provides a significant improvement in OA, as shown in Table V. The influence of the two modules on the two H$^2$ datasets is not significantly different. The plot distribution for the WHU-Hi-LongKou is more regular, so the S$^2$AiA module focusing on global spatial–spectral feature extraction can achieve better results. The WHU-Hi-HongHu has various plot sizes and many fragmented plots, so the MSA module focusing on MSA can achieve better results. Similarly, on Pavia University, both modules result in an improved network performance, but the S$^2$AiA module achieves better results. Therefore, to enable the network to use global spatial–spectral information while also considering local multiscale features, we add both the S$^2$AiA module and MSA module at the same time. For visual evaluation, we also present visualization maps of the proposed

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TABLE III
CLASSIFICATION ACCURACIES (%) FOR THE WHU-Hi-HongHu Dataset

| No. | SVM | R-PCACNN | HybridSN | SSFTT | FPGA | S$^3$ANet | FullyContNet | S$^2$AiANet |
|---|---|---|---|---|---|---|---|---|
| C1 | 85.22 | 85.41 | 97.34 | 95.29 | 96.15 | 97.43 | 95.01 | 97.07 |
| C2 | 71.81 | 72.41 | 87.27 | 92.17 | 92.14 | 94.87 | 88.24 | 96.01 |
| C3 | 71.73 | 68.83 | 93.49 | 91.17 | 89.64 | 93.67 | 97.66 | 95.65 |
| C4 | 75.99 | 82.34 | 96.20 | 94.09 | 96.26 | 96.83 | 96.73 | 98.69 |
| C5 | 59.76 | 79.11 | 97.79 | 95.55 | 98.81 | 99.26 | 99.82 | 99.52 |
| C6 | 83.22 | 87.15 | 95.36 | 94.92 | 92.64 | 94.74 | 90.15 | 96.10 |
| C7 | 49.64 | 52.66 | 63.23 | 72.33 | 71.32 | 76.49 | 73.26 | 78.59 |
| C8 | 25.19 | 49.86 | 82.53 | 90.02 | 91.19 | 98.56 | 99.08 | 97.69 |
| C9 | 83.69 | 91.41 | 99.25 | 95.96 | 97.21 | 99.56 | 97.04 | 98.04 |
| C10 | 42.06 | 51.86 | 82.29 | 82.97 | 81.49 | 83.67 | 78.68 | 88.12 |
| C11 | 37.23 | 43.76 | 54.25 | 81.96 | 90.36 | 83.89 | 90.13 | 89.15 |
| C12 | 42.99 | 43.51 | 69.52 | 68.75 | 84.07 | 86.96 | 93.28 | 87.86 |
| C13 | 39.25 | 50.72 | 72.06 | 74.58 | 78.88 | 74.04 | 82.95 | 85.82 |
| C14 | 55.83 | 62.47 | 84.35 | 92.29 | 91.69 | 92.76 | 88.25 | 91.39 |
| C15 | 79.32 | 87.21 | 98.36 | 99.89 | 97.85 | 97.65 | 96.62 | 97.44 |
| C16 | 77.38 | 82.41 | 95.23 | 97.44 | 91.03 | 94.75 | 95.74 | 97.00 |
| C17 | 50.42 | 71.52 | 99.97 | 99.32 | 98.63 | 98.69 | 99.93 | 99.13 |
| C18 | 65.76 | 82.89 | 99.25 | 99.59 | 98.50 | 99.15 | 99.62 | 97.87 |
| C19 | 58.18 | 73.25 | 81.27 | 89.01 | 92.54 | 87.77 | 80.53 | 92.54 |
| C20 | 67.96 | 64.49 | 97.60 | 92.48 | 97.83 | 97.37 | 99.50 | 98.79 |
| C21 | 60.32 | 75.52 | 100.00 | 100.00 | 100.00 | 99.92 | 99.61 | 99.77 |
| C22 | 69.69 | 79.05 | 88.79 | 92.60 | 99.88 | 99.95 | 99.77 | 98.06 |
| OA | 68.35 | 74.49 | 89.59 | 90.31 | 91.79 | 92.70 | 92.43 | **94.94** |
| AA | 61.48 | 69.90 | 87.97 | 90.56 | 92.19 | 93.09 | 92.80 | **94.56** |
| Kappa | 0.6196 | 0.6899 | 0.8695 | 0.8791 | 0.8971 | 0.9083 | 0.9051 | **0.9363** |

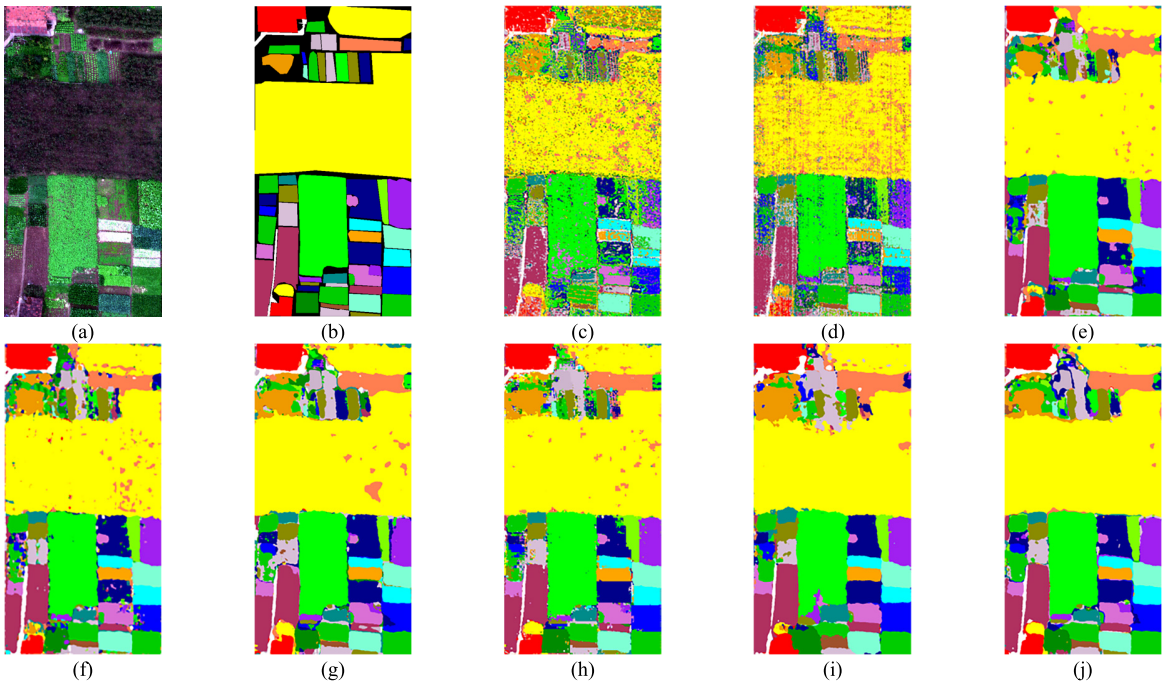


Fig. 9. Classification results for the WHU-Hi-HongHu dataset. (a) False-color image. (b) Ground-truth image. (c) SVM (OA = 68.35%). (d) R-PCACNN (OA = 74.49%). (e) HybridSN (OA = 89.59%). (f) SSFTT (OA = 90.31%). (g) FPGA (OA = 91.79%). (h) S$^3$ANet (OA = 92.70%). (i) FullyContNet (OA = 92.43%). (j) S$^2$AiANet (OA = 94.94%).

module on the WHU-HI-HongHu dataset. The visualization maps shown in Fig. 11 also show the same results. The use of a single module reduces the part of the middle cotton area that is wrongly divided into cotton firewood. When used simultaneously, there is almost no misclassification in the middle area. The results obtained on the above three datasets show that, when the two modules are added at the same time, the network achieves superior classification accuracy.

### G. Exploration of the Attention Mechanism

In this section, we describe the experiment conducted to explore the use of the attention mechanism in the network. To prevent gradient explosion, we adjusted the learning rate to 0.0005 and the weight decay to 0.0001. The encoder–decoder structure of the proposed S$^2$AiANet network served as the backbone for the experiment. First, the impact of using an attention mechanism and an attention-in-attention mechanism on the network performance was investigated. Based on the

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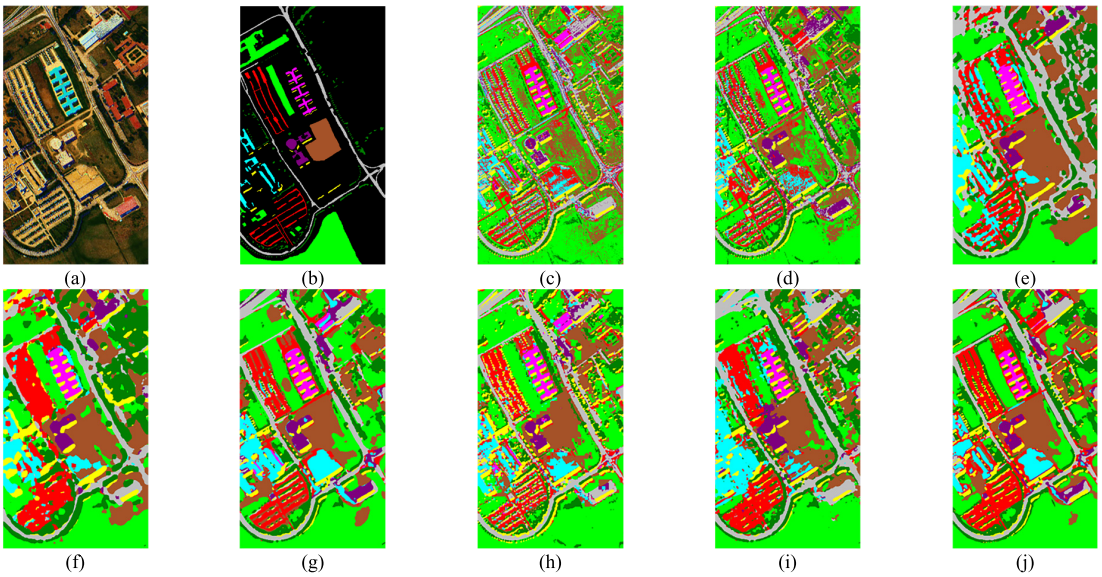

Fig. 10. Classification results for the Pavia University dataset. (a) False-color image. (b) Ground-truth image. (c) SVM (OA = 82.56%). (d) R-PCACNN (OA = 84.52%). (e) HybridSN (OA = 92.11%). (f) SSFTT (OA = 92.59%) (g) FPGA (OA = 94.24%). (h) $S^3$ANet (OA = 94.28%). (i) FullyContNet (OA = 96.14%). (j) $S^2$AiANet (OA = 97.37%).

TABLE IV

CLASSIFICATION ACCURACIES (%) FOR THE PAVIA UNIVERSITY DATASET

| No. | SVM | R-PCACNN | HybridSN | SSFTT | FPGA | $S^3$ANet | FullyContNet | $S^2$AiANet |
|---|---|---|---|---|---|---|---|---|
| C1 | 73.07 | 76.7 | 88.54 | 82.13 | 83.88 | 88.95 | 93.98 | 88.36 |
| C2 | 87.31 | 93.13 | 93.15 | 96.47 | 95.27 | 95.00 | 95.99 | 99.88 |
| C3 | 78.78 | 75.55 | 97.59 | 92.09 | 94.50 | 90.26 | 98.26 | 90.69 |
| C4 | 82.63 | 90.23 | 88.15 | 90.75 | 93.91 | 94.77 | 97.14 | 96.35 |
| C5 | 99.32 | 99.85 | 99.85 | 99.62 | 100.00 | 100.00 | 100.00 | 99.92 |
| C6 | 72.62 | 57.79 | 99.78 | 93.06 | 97.40 | 99.38 | 96.82 | 100.00 |
| C7 | 84.52 | 98.39 | 99.39 | 99.84 | 99.39 | 98.31 | 97.93 | 99.77 |
| C8 | 80.04 | 77.41 | 76.65 | 86.24 | 98.22 | 90.24 | 94.80 | 99.43 |
| C9 | 99.78 | 99.78 | 95.55 | 98.59 | 99.89 | 100.00 | 100.00 | 100.00 |
| OA | 82.56 | 84.52 | 92.11 | 92.59 | 94.24 | 94.28 | 96.14 | **97.37** |
| AA | 84.23 | 85.42 | 93.18 | 93.20 | 95.83 | 95.21 | **97.21** | 97.16 |
| Kappa | 0.7715 | 0.7937 | 0.8966 | 0.9022 | 0.9243 | 0.9250 | 0.9491 | **0.9651** |

TABLE V

EFFECTIVENESS ANALYSIS (OA%) FOR THE PROPOSED MODULES

| Method | LongKou | HongHu | PaviaU |
|---|---|---|---|
| FreeNet | 96.06 | 91.79 | 94.24 |
| FreeNet+MSA | 97.65 | 94.07 | 94.49 |
| FreeNet+$S^2$AiA | 97.89 | 93.99 | 95.16 |
| FreeNet+MSA/$S^2$AiA | 98.45 | 94.25 | 96.37 |

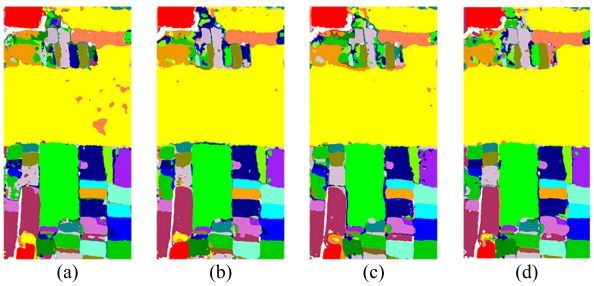

Fig. 11. Visual display of the proposed module on the WHU-HI-HongHu dataset. (a) FreeNet. (b) FreeNet+MSA. (c) FreeNet+$S^2$AiA. (d) FreeNet+MSA/$S^2$AiA.

proposed structure of the $S^2$AiA module, the attention mechanism uses traditional convolution operations to obtain the

TABLE VI

ACCURACY (OA%) FOR DIFFERENT ATTENTION MECHANISMS

| Method | LongKou | HongHu | PaviaU |
|---|---|---|---|
| Backbone+A (parallel connection) | 96.88 | 91.30 | 95.89 |
| Backbone+AiA (parallel connection) | 97.97 | 92.69 | 96.71 |
| Backbone+$A_{spa}iA_{spe}$ (series connection) | 93.43 | 89.74 | 88.62 |
| Backbone+$A_{spe}iA_{spa}$ (series connection) | 94.94 | 88.37 | 88.19 |

values of $Q$ and $K$, while the attention-in-attention mechanism uses attention mechanisms to obtain the values of $Q$ and $K$. As shown in Table VI, "Backbone+A" represents using traditional attention and "Backbone+AiA" represents using attention in attention. The results obtained from three publicly available datasets show that using the attention-in-attention mechanism can better utilize the global spatial–spectral information and achieve superior classification results. In addition, the effect of the series connection structure was also investigated. We studied two different sequential structures: as shown in Table VI, "Backbone+$A_{spa}iA_{spe}$" represents extracting spatial features first, and then, spectral features and

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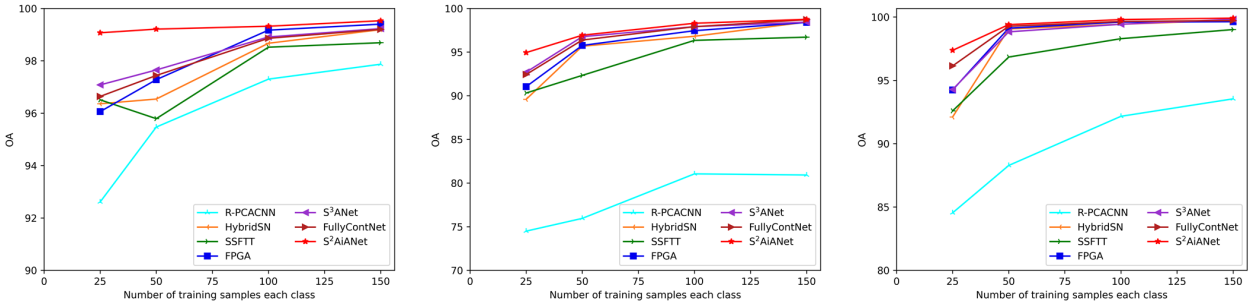
TABLE VII
COMPUTATIONAL EFFICIENCY AND COMPLEXITY OF DIFFERENT METHODS ON THE WHU-HI-HongHu Dataset

| | R-PCACNN | HybridSN | SSFTT | FPGA | S³ANet | FullyContNet | S²AiANet |
|---|---|---|---|---|---|---|---|
| **Test time (S)** | 18.95 | 132.99 | 61.48 | 1.15 | 1.25 | 1.16 | 1.25 |
| **Parameters (M)** | 29.39 | 512.32 | 14.89 | 269.16 | 504.49 | 140.92 | 405.52 |
| **GFLOPs** | 0.0005 | 0.2495 | 0.0169 | 231.75 | 175.44 | 213.12 | 187.32 |




(a) (b) (c)

Fig. 12. Effect of different numbers of training samples for the methods based on deep learning. (a) WHU-Hi-LongKou dataset. (b) WHU-Hi-HongHu dataset. (c) Pavia University dataset.

TABLE VIII
EFFECTIVENESS ANALYSIS OF MASF MODULES WITH DIFFERENT NUMBERS OF SCALES

| | 1 | 2 | 4 | 8 |
|---|---|---|---|---|
| **HongHu** | 93.31 | 93.35 | 94.07 | 93.00 |

"Backbone+$A_{spe}iA_{spa}$" represent extracting spectral features first and then spatial features. It is not difficult to find that the accuracy of parallel connection is higher than that of series connection. Because the series connection method weakens the feature maps extracted from the first half of the module, it emphasizes the feature maps extracted from the second half of the module. This makes the network focus either on spectral features or spatial features, limiting the improvement of the network accuracy. Therefore, a parallel-structured S²AiA module is proposed to enhance the network's ability to extract global spatial–spectral features and further improve the network performance.

### H. Computational Efficiency and Complexity

In addition to accuracy, computational efficiency and complexity are also important indicators for measuring the quality of deep learning methods, which can reflect the potential of classification methods in practical applications. Table VII shows test time, parameters, and GFLOPs for different classification methods on the WHU-HI-HongHu dataset. All the deep learning methods are implemented in PyTorch, and the network structure is implemented through the nn.Module. They are run in the same configuration, with the Tesla P100-PCIE-16GB GPU and Intel[1] Xeon[1] CPU E5-2690 v4 CPU. The Patch-based methods are pixel-by-pixel prediction, which is influenced by the number of pixels to be predicted and the patch size. The HybridSN with the largest selected patch size clearly achieved the slowest testing time on the WHU-HI-HongHu dataset. The FCN-based methods take a very short time because they are end-to-end predictions, just over a second. In addition, to better evaluate the effectiveness of various methods and help researchers choose appropriate methods in different situations, we also calculated the parameters and GFLOPs of various methods on the WHU-HI-HongHu dataset for reference.

[1]Registered trademark.

### I. Sensitivity Analysis of the Number of Scales

The selection of the number of scales in multiscale modules is very important and should be discussed and analyzed. Similarly, we selected the FreeNet of FPGA as a baseline and conducted comparative experiments by using MASF modules with scales number 1 (rate = 7), 2(rate = 3 and 12), 4(rate = 3, 6, 9, and 12), 8 (rate = 3, 4, 5, 6, 9, 10, 11, and 12). To ensure that the range of the receptive field remains constant, we have limited the rate (from 3 to 12) and changed the number of scales between them. We chose the WHU-HI-HongHu dataset for the experiment, which has complex scenes and big differences in scales. The results are shown in Table VII. It is not difficult to see that as the number of scales increases, the accuracy increases, but when the number of scales increases from 4 to 8, the accuracy decreases. We analyze this because as the number of scales increases, the number of feature channels (256/number) in feature maps at different scales decreases. It will reduce the expression ability of features at different scales, resulting in a decrease in accuracy. Therefore, we chose the four scales (rate = 3, 6, 9, and 12) to achieve the best classification results.

### J. Sensitivity in Relation to Training Set Size

Deep learning is a data-driven approach, and the size of the training set has a significant impact on the accuracy of the model. Therefore, we analyzed the impact of different sizes of training sets on model accuracy with two H² datasets and one classic HSI dataset. The number of training labeled samples varies from 25 to 150 per class over three datasets. The specific OAs are shown in Fig. 12. It is evident that with the increasing number of training samples, the accuracy continues to improve, and except for the SSFTT method, it is OA has a slight decrease in training samples of 50 per

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class in the WHU-Hi LongKou dataset. In addition, it is not difficult to find that, when the number of training samples reaches 100 or more per class, the FCN-based methods can achieve excellent classification accuracy. Also, the proposed $S^2$AiANet has always achieved the best classification accuracy. However, when the training samples are small, especially 25 in each class, the $S^2$AiANet method exhibits a more competitive performance than other comparison methods.

## V. Conclusion

In this article, the $S^2$AiANet method has been proposed for HSI classification. By introducing the $S^2$AiA module when calculating the spatial attention map and spectral attention map, the different weights of the different bands and positions are fully considered, which helps the network to further extract the global spatial–spectral context information and enhances the modeling ability of the network for long-range dependence relationships between pixels. Even when the number of training samples is greatly reduced, the network can still extract features with sufficient discriminative ability and obtain a high classification accuracy. In addition, in the task of fine classification of crops, there can be many kinds of crops and the planting areas of various crops can also be different. Therefore, to enhance the robustness of the network under different datasets, the MSA module is proposed, which directly weights the features of different scales to improve the classification accuracy of the network in the face of different complex scenes. Extensive experiments conducted on multiple datasets showed that the proposed $S^2$AiANet method can achieve a superior classification performance, compared with the state-of-the-art hyperspectral classifiers.

## References


[1] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution ($H^2$) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[2] S. Jakob, R. Zimmermann, and R. Gloaguen, "The need for accurate geometric and radiometric corrections of drone-borne hyperspectral data for mineral exploration: MEPHySTo—A toolbox for pre-processing drone-borne hyperspectral data," *Remote Sens.*, vol. 9, no. 1, p. 88, Jan. 2017.

[3] O. Nevalainen et al., "Individual tree detection and classification with UAV-based photogrammetric point clouds and hyperspectral imaging," *Remote Sens.*, vol. 9, no. 3, p. 185, Feb. 2017.

[4] T. T. Sankey, J. McVay, T. L. Swetnam, M. P. McClaran, P. Heilman, and M. Nichols, "UAV hyperspectral and LiDAR data and their fusion for arid and semi-arid land vegetation monitoring," *Remote Sens. Ecology Conservation*, vol. 4, no. 1, pp. 20–33, Mar. 2018.

[5] H. Zhao et al., "Mapping the distribution of invasive tree species using deep one-class classification in the tropical montane landscape of Kenya," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 328–344, May 2022.

[6] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023.

[7] J. Yuan, S. Wang, C. Wu, and Y. Xu, "Fine-grained classification of urban functional zones and landscape pattern analysis using hyperspectral satellite imagery: A case study of Wuhan," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3972–3991, 2022.

[8] P. Duan, X. Kang, P. Ghamisi, and S. Li, "Hyperspectral remote sensing benchmark database for oil spill detection with an isolation forest-guided unsupervised detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–11, 2023.

[9] P. Singh et al., "Hyperspectral remote sensing in precision agriculture: Present status, challenges, and future trends," in *Hyperspectral Remote Sensing*. Amsterdam, The Netherlands: Elsevier, 2020, pp. 121–146.

[10] H. Zhao, Y. Zhong, X. Wang, and H. Shu, "One-class risk estimation for one-class hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023.

[11] L. P. Pott, T. J. C. Amado, R. A. Schwalbert, G. M. Corassa, and I. A. Ciampitti, "Satellite-based data fusion crop type classification and mapping in Rio Grande do Sul, Brazil," *ISPRS J. Photogramm. Remote Sens.*, vol. 176, pp. 196–210, Jun. 2021.

[12] F. Löw, U. Michel, S. Dech, and C. Conrad, "Impact of feature selection on the accuracy and spatial uncertainty of per-field crop classification using support vector machines," *ISPRS J. Photogramm. Remote Sens.*, vol. 85, pp. 102–119, Nov. 2013.

[13] Y. Zhong et al., "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[14] C. Zhang and J. M. Kovacs, "The application of small unmanned aerial systems for precision agriculture: A review," *Precis. Agricult.*, vol. 13, no. 6, pp. 693–712, Dec. 2012.

[15] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2012.

[16] B. Krishnapuram, L. Carin, M. A. T. Figueiredo, and A. J. Hartemink, "Sparse multinomial logistic regression: Fast algorithms and generalization bounds," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 6, pp. 957–968, Jun. 2005.

[17] G. M. Foody and A. Mathur, "A relative evaluation of multiclass image classification by support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 6, pp. 1335–1343, Jun. 2004.

[18] J. Xia, L. Bombrun, T. Adali, Y. Berthoumieu, and C. Germain, "Spectral–spatial classification of hyperspectral images using ICA and edge-preserving filter via an ensemble strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4971–4982, Aug. 2016.

[19] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[20] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[21] X. Hu et al., "SPNet: Spectral patching end-to-end classification network for UAV-borne hyperspectral imagery with high spatial and spectral resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[22] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral–spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.

[23] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5612–5626, Aug. 2020.

[24] S. Jia and Y. Wang, "Multiscale convolutional transformer with center mask pretraining for hyperspectral image classification," 2022, *arXiv:2203.04771*.

[25] J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and J. Li, "Visual attention-driven hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8065–8080, Oct. 2019.

[26] F. Wang et al., "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6450–6458.

[27] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[28] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[29] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:43:10 UTC from IEEE Xplore. Restrictions apply.

[30] X. Hu, X. Wang, Y. Zhong, and L. Zhang, "$S^3$ANet: Spectral–spatial-scale attention network for end-to-end precise crop classification based on UAV-borne $H^2$ imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 147–163, Jan. 2022.
[31] W. Guo, H. Ye, and F. Cao, "Feature-grouped network with spectral–spatial connected attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.
[32] L. Fang, Z. Liu, and W. Song, "Deep hashing neural networks for hyperspectral image feature extraction," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1412–1416, Sep. 2019.
[33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[34] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
[35] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.
[36] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023.
[37] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–20, 2023.
[38] D. Wang, B. Du, and L. Zhang, "Fully contextual network for hyperspectral scene parsing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.
[39] Y. Shen et al., "Efficient deep learning of nonlocal features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6029–6043, Jul. 2021.
[40] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.
[41] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
[42] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[43] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.
[44] A. Borji and L. Itti, "State-of-the-art in visual attention modeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 185–207, Jan. 2013.
[45] M. Carrasco, "Visual attention: The past 25 years," *Vis. Res.*, vol. 51, no. 13, pp. 1484–1525, Jul. 2011.
[46] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[47] X. Wang, K. Tan, P. Du, C. Pan, and J. Ding, "A unified multiscale learning framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022.
[48] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

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